

sufficient mining power, it will win. Forking is a key mechanism to assure efficiency in *DeFi*.

Gas. A fee required to execute a transaction and to execute a *smart contract*. The mechanism that allows *Ethereum* to deal with the *halting problem*.

Geoblock. Technology that blocks users from certain countries bound by regulation that precludes the application.

Governance token. The right of an owner to vote on changes to the protocol. Examples include the MakerDAO MKR token and the Compound COMP token.

Halting problem. A computer program in an infinite loop. *Ethereum* solves this problem by requiring a fee for a certain amount of computing. If the *gas* is exhausted, the program stops.

Hash. See *cryptographic hash*.

Hexadecimal. A counting system in base-16 that includes the first 10 numbers 0 through 9 plus the first six letters of the alphabet, a through f. Each hexadecimal character represents 4 bits, where 0 is 0000 and the 16th (f) is 1111.

Horizontal scaling. An approach that divides the work of the system into multiple pieces, retaining decentralization but increasing the throughput of the system through parallelization. Also known as *sharding*. *Ethereum 2.0* takes this approach in combination with a *proof-of-stake* consensus algorithm.